

BY 510: Advanced Cell Biology

Course Code	BY 510
Course Name	Advanced Cell Biology
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	UG/PG
Distribution	Elective
Approval	9th Senate (May 2016)

Course Contents

- **A tour to Cell**
Exploring Eukaryotic and plant cell; cell components, their structure and function; cytoskeleton network: components and structural functions; Cell-Cell and cellmatrix adhesion; Extracellular Matrix, Plant cell wall, Animal cell membrane; Cell junctions, plasmodesmata, gap junctions, desmosomes and tight junction [8 Lectures]
- **Membrane structure and function**
Membrane models; evolution of different membrane lipids, Membrane proteins and their functions, Membrane carbohydrates and their roles in cell-cell recognition; The permeability of the bi-layers: transport proteins; Passive, active and co transport by antiporters and symporters, transporter proteins in plant vacuoles exocytosis and endocytosis. [8 Lectures]
- **Cell signalling and transport across cell membranes**
Signalling molecules and cell surface receptors; intracellular signal transduction; G protein coupled receptors; plant growth factors and hormones-auxins, gibberlines, cytokines and others passive and active transport; transport into prokaryotic cells; endocytosis, exocytosis; entry of viruses and toxins into cells [8 Lectures]
- **Membrane trafficking**
Translocation of secretory proteins across the ER membrane; protein modifications, folding and quality control in the ER; export and sorting of proteins to mitochondria, chloroplast and peroxisomes. [8 Lectures]
- **Eukaryotic cell cycle**
Biochemical and genetics studies on cell cycle; mechanisms regulating mitotic events; meiosis - a special type of cell division; Cell Birth, lineage and 2 death: Asymmetrical cell division, patterns of stem cell division; Biological description of apoptosis; Molecular basis of cancer, oncogenes and tumor suppressor genes [10 Lectures]

Text and Reference Books

1. Lodish et al., **Molecular Cell Biology**, 5th edition or later, W.H. Freeman and Company.
2. Alberts et al., **Molecular Biology of the Cell**, 4th edition or later, Garland Sciences.
3. by David L. Nelson, Michael M. Cox, **Lehninger Principles of Biochemistry**, 6th Edition.
4. L. Stryer, **Biochemistry**, 4th edition, W. H. Freeman & Company.
5. Additional material from recent scientific papers/reviews will be provided